

passed away, for instance, could prove undeniably useful for the ensuing investigation.

Determining Time of Death

After someone dies, the body goes through several changes as it nears decomposition. When determining the time of death, or the **Post Mortem Interval (PMI)**, for any given individual, it is important to take all possible effects into account. Where was the body when found? When was the body found? Did the decedent have any previous health conditions that could affect postmortem readings?

With so many questions and more, it is impossible to definitively state the exact time of death without some external record such as witnesses or timed footage. However, the medical examiner can approximate a time of death within reasonable accuracy with what is apparent and can then narrow or widen that window based on new information as it arises. In many cases, the examiner will be presented with a body in the **Early PMI**, which is classified as between 3 and 72 hours after death. In Early PMI cases, examiners will estimate the time of death using the following postmortem physical changes: rigor mortis, algor mortis, and livor mortis.

Synthesis

The coroner or medical examiner will use as many of the following observations as possible to develop a theory for the time of death: rigor mortis, algor mortis, livor mortis, degree of putrefaction, stomach contents, cloudiness of the corneas, vitreous fluid, insect activity, and any other information gathered from the scene.

Rigor Mortis

Rigor mortis is the stiffening of the body's muscles from the gradual depletion of adenosine triphosphate (ATP), the organic compound responsible for breaking down the actin-myosin filaments that form naturally during the process of muscle contraction. The first signs of rigor mortis will appear in the small muscles of the face, such as the eyes and mouth, roughly 2 to 6 hours after death. From there, the progression continues into the rest of the body, including the internal organs, reaching its maximum stiffness around 12 hours after death. After full onset, the body remains stiff until roughly 24 hours after death, when the actin-myosin complex that has formed starts disintegrating. This process happens simultaneously across the body but is also first perceptible in the smaller muscles of the face. At around 36 hours postmortem, the stiffness of rigor mortis disappears completely, giving way to the phase known as **secondary flaccidity**. Since rigor mortis eventually disappears, it is called **transient evidence**.

Algor Mortis

The change in body temperature after death, also known as **algor mortis**, is one of the many factors in estimating the time a patient died and one of the simpler early determiners. While there are many elements that affect body temperature postmortem — such as the location of the body — it is one of the first metrics recorded by an emergency responder.

Similarly, the temperature of the location is extremely important to a medical examiner as the body slowly conforms to the environment around it. Determining whether the surroundings are warmer or cooler than the body's natural antemortem temperature can give a better understanding of how long the body has been deceased, assuming a rate of ± 1.5 degrees per hour. While this formula is not exact, it is extremely helpful in determining a **preliminary** estimate of time of death before further analysis can be conducted.



Figure 4.2 The temperature of a body is affected by ambient temperature.

The temperature of the surrounding area will determine whether a body cools or warms after death. For instance, if a body was found in the trunk of a car on the hottest day on record in Texas, we would know that the body's temperature would rise from the moment of death until the body was found or reached the surrounding ambient temperature (given that the body was not moved and the temperature remained relatively consistent). However, if the body was found in an icy river in Michigan, the body would be expected to cool before being found. Essentially, whether a body warms or cools after death is primarily reliant upon one variable: whether the temperature of surrounding air is warmer or cooler than the body's temperature before death.



The **average temperature** of the human body while living is around 98.6 degrees Fahrenheit. If given a hypothetical corpse that was found to be roughly 79 degrees at 4:32 PM, the medical examiner, in order to determine the number of hours passed since death, would find the difference between the two temperatures divided by the rate of postmortem temperature change.

In this case, $98.6 - 79 = 19.6$, indicating the body has lost 19.6 degrees. Divide 19.6 by the standard rate of change of 1.5, and the result is 13.07.

In this example, this would mean the decedent would have passed away approximately 13 hours before being found, sometime around 3:32 AM. As a best practice, the coroner or medical examiner would then take that approximation and add 30 minutes to each side as a margin of error. Thus, the final preliminary window for time of death is 03:02 – 04:02.

If air temperature is lower than body temperature:

$$[\text{POST MORTEM INTERVAL}] = [\text{TIME FOUND}] - \left(\frac{98.6 - [\text{BODY TEMPERATURE WHEN FOUND}]}{1.5} \right)$$

If air temperature is higher than body temperature:

$$[\text{POST MORTEM INTERVAL}] = [\text{TIME FOUND}] - \left(\frac{[\text{BODY TEMPERATURE WHEN FOUND}] - 98.6}{1.5} \right)$$

Livor Mortis

The final common change in the Early PMI is **livor mortis**, a blueish/purplish discoloration of the skin following death due to the pooling of blood. While it cannot be seen with the naked eye at first, livor mortis begins 20 to 30 minutes after death. After 3 to 6 hours, dark spots appear in larger patches, with maximum lividity occurring 6 to 12 hours after death. The discoloration becomes “fixed” after this point, which can be confirmed by applying pressure. Fixation has traditionally been used to denote a PMI greater than 12 hours, although this technique has fallen out of favor in recent years.

Remember:

Time of death windows are recorded in the format of HH:MM – HH:MM, with the earliest possible time written first.